

Does the claimed subquadratic 3SUM transfer to index calculus?

Date: 2026-10-07. Status: analysis (no code changes, no new measurements). Base commit: 6350e7f7 (origin/main). Location: `experiments/threesum-ic-transfer-20261007/`. Visual: `transfer.dot` (transfer-breakage diagram), rendered to `figures/transfer.svg` and `figures/transfer.pdf`. PDF of this report: `REPORT-2026-10-07.pdf` (built from this file; contains the figure).

Claim labels: **[reported]** (in the cited preprint/aggregator, not verified here), **[structural]** (verifiable in the cited source), **[derived]** (follows from cited facts), **[proposed]** (future work only). Short answer: **no transfer**. Three independent gaps block it, and no IC row or floor moves. The falsification checklist in §5 says exactly what would have to change for that verdict to flip.

1. Question, scope, method

Question. Alman and Vassilevska Williams posted arXiv 2610.06783 on Oct 5, 2026, claiming deterministic $O(n^{1.9992})$ 3SUM and thereby refuting the 3SUM hypothesis. Does anything in this repository’s index-calculus program get enabled by, or need revision because of, that claim?

Scope. The claim as reported on Oct 6-7 (preprint + aggregator coverage); this repo’s IC decomposition oracles, cost ledgers, and lower bounds; the sibling `crypto` repo checked for 3SUM-hypothesis dependencies. Out of scope: adjudicating the preprint’s correctness (unreviewed; see §6).

Method. Read the arXiv abstract/record for the claimed statements; grepped both repos for 3SUM/3-sum/fine-grained dependencies; compared the 3SUM query model against the IC oracles’ query model primitive by primitive. No implementation of the claimed algorithm was attempted (see Gap 3 for why that is the correct call).

2. The claim as reported

- Deterministic 3SUM on n polynomial-size integers in $O(n^{1.9992})$, and integer-weight APSP in $O(n^{2.9995})$ **[reported]** (arXiv 2610.06783).
- Technique: fast selected entries of thin matrix products (a variant of Coppersmith’s rectangular matrix multiplication from a Schönhage ten-multiplication identity), interpreted as sparse-triangle finding on lopsided graphs, reached by known reductions **[reported]** (same source).
- A Lean 4 certificate of the main theorems is reported by secondary aggregators only; the paper’s own AI-disclosure and formalization status could not be verified here and conflicting aggregator claims exist, so all of that stays **[reported]** rather than cited fact.
- Corollaries claimed: refutations of the real-valued 3SUM/APSP hypotheses, Exact Triangle, Zero-Weight k -Clique, and three hinted OMv conjectures **[reported]**.

3. Transfer analysis: three independent gaps

The tempting analogy is real: IC relation collection searches for m -tuples summing to a target, which smells like k -SUM. The transfer fails at each of three links; any one of them suffices.

Gap 1 – model [structural]. The claimed algorithm decides 3SUM over ordered integers (comparisons, exact triangles). IC decomposition works over elliptic-curve groups (F_{2^n}, F_p) accessed by hash lookups, with lifting/twist/cofactor structure the integer problem has no analogue of. Moreover the regime is inverted: 3SUM is interesting with $O(n)$ space, while IC pays $O(|F|^2)$ space for a pair table once and then answers each query in $O(1)$ lookups (`suite/docs/ic/README.md`, pair-table oracle). Subquadratic decision buys nothing where the incumbent already pays quadratic memory for constant-time queries.

Gap 2 – task [structural]. 3SUM is one-shot decision: does a triple exist? IC oracles must *enumerate witnesses* (the actual summands), repeatedly, over thousands of targets sharing one table, with per-target verification by re-addition. A decision procedure supplies none of: witness recovery, enumeration over a target stream, or the lift checks (`lift_failures`, `unliftable_systems`) that the accounting contract prices (`suite/docs/ic/FRAMEWORK.md` §3).

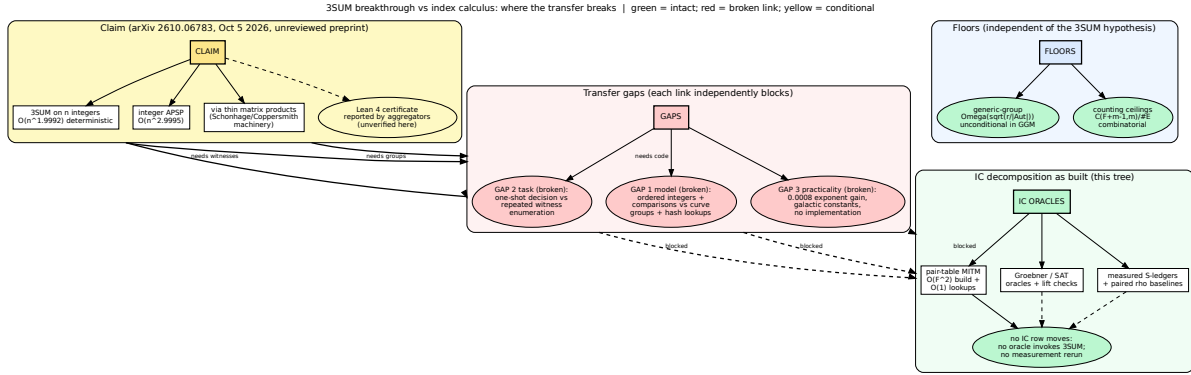


Figure 1: Transfer-breakage diagram. Each of the three gaps independently blocks the claimed transfer; floors stand on independent ground. See the SVG for zoomable text.

Gap 3 – practicality [derived]. The exponent gain is 0.0008, through rectangular-MM machinery with galactic constants and no implementation. At every feasible size a hash table wins by orders of magnitude; there is no code to benchmark, so no measured comparison can even be staged. Attempting an implementation here would be pure cost with no falsifiable upside.

Query-model comparison (why the analogy fails):

aspect	3SUM (claimed)	IC pair-table oracle	IC algebraic oracle
domain	ordered integers	curve group + hashes	Boolean/ $\mathbb{F}_p$ systems
task	one-shot decision	repeated witness lookup	witness + lift check
preprocessing	none stated	$O(F^2)$ table, amortized	per-query system build
per-query cost	$O(n^{1.9992})$ total	$O(1)$ -ish lookups	Gröbner/SAT + lift
implementation	none exists	measured S -ledgers	measured S -ledgers

4. Floors and ledgers: nothing moves

- Generic-group lower bounds ($\Omega(\sqrt{r/|Aut|})$, Shoup/Nechaev) are unconditional in the generic group model, not 3SUM-conditional [structural]. A 3SUM refutation cannot touch them.
- Counting ceilings $C(F+m-1, m)/\#E$ and the exact cofactor-class refinements) are combinatorial identities over the experiment's own enumeration [structural].
- Repo `3sum` strings are unrelated: `dense_3sum_*wall_bits` fields in `crypto` boundary-target fixtures are 3-summand dense-oracle budgets, and `b3sum` hits are the BLAKE3 tool; no file in either repo invokes the 3SUM problem or hypothesis (grep over both trees, Oct 7).
- No IC oracle invokes a 3SUM subroutine, so no measured row, paired rho baseline, or verdict changes. This note is an accounting entry, not a result.

5. What would flip the verdict (falsification checklist + pending protocol)

The verdict flips only if ALL of these hold; each is independently checkable:

1. The preprint (or a successor) survives peer review, or its Lean certificate is independently reproduced from the repo's own checkout.
2. A concrete k-SUM-style oracle exists for *group* sums with hash access that returns witnesses, not just decisions.
3. That oracle is implemented (any language) with stated constants.

4. On a frozen one-target workload with the repo’s accounting contract, the oracle’s whole-pipeline S (table build included, failures included, verified answers only) beats the pair-table reference row on the same base, with the paired rho baseline re-run.

Preregistered protocol (pending, not run): `ic bench matrix pairing mitm` (reference) against one `descent-ksum` candidate oracle on the same `binary-subspace` base, `walk` targets, and seeds, plus the mandatory `correct: YES` + matched-rho gate per `FRAMEWORK.md` §7. The candidate cell cannot be filled until checklist items 1-3 hold; running the reference alone would only re-measure the known row. If a future 3SUM-derived oracle arrives, this section becomes its run sheet.

6. Negative findings and uncertainty

1. No transfer path was found; the three gaps above are each sufficient to block it.
2. No repo content depends on the 3SUM hypothesis, so the hypothesis’s possible fall changes no in-repo conclusion.
3. Uncertainty is concentrated in §2: preprint unreviewed, aggregator claims about AI discovery and Lean verification conflict and are unverified here. Nothing in §§3-5 depends on them either way.
4. The sibling-repo grep covered `research/`; a full-corpus fine-grained survey was not run, but the floors that matter are cited to their unconditional sources, not to an absence claim.

7. Canonical graphs checked (why they remain current)

This note asserts no new curve, edge, base, run, or speedup. Checked and unchanged: `experiments/ic-assumption-graph/` (no assumption status moves: no oracle changes, no floor changes); `experiments/ec-arithmetic-optimizations/` (no technique added or repriced); `experiments/volcano-ic/figures/`, `experiments/ic-bench/` baselines, `paper/` figures; `curves.yaml`, `isogeny_routes.json`; `fb-archive/index.csv` (this directory matches no workflow path filter).

8. Reproduce

```
cd experiments/threesum-ic-transfer-20261007
dot -Tsvg transfer.dot -o figures/transfer.svg
dot -Tpdf transfer.dot -o figures/transfer.pdf
pandoc REPORT-2026-10-07.md -o REPORT-2026-10-07.pdf \
  --pdf-engine=pdflatex -V geometry:margin=1in -V colorlinks:true
```

Graphviz 2.43.0, pandoc 3.1.3, pdfTeX 3.141592653-2.6-1.40.25 (TeX Live 2023/Debian) on this VM. No quantitative figure: nothing was measured, and a chart of one unimplemented exponent against measured oracles would imply a comparison that does not exist.

9. References

- arXiv 2610.06783 (Alman, Vassilevska Williams), abstract/record as served Oct 6-7 2026 [**reported**].
- Aggregator coverage (MindPattern, Post-Cutoff, ArXivSignals), Oct 5-6 2026 [**reported, unverified here**].
- `suite/docs/ic/README.md` (pair-table oracle, GAE pinning), `suite/docs/ic/FRAMEWORK.md` §3-4, §7 (lift accounting, oracle contract, reporting rules), `suite/docs/ECDLP_ATTACK_MATRIX.md` §10b (phase-split verdicts) – all at base commit `6350e7f7`.
- `crypto` boundary-target fixtures (`dense_3sum_*wall_bits` fields) for the negative grep result.